

# MEDICAL DEVICE MARKETING

## UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

### 1. Project Definition

#### Introduction

A **project** is a planned set of activities carried out to achieve a specific objective within a defined **scope, time, cost, and quality**.

In medical device development, a project may involve designing, developing, testing, validating, obtaining regulatory clearance/approval, and launching a new medical device.

#### Example

Development of an **IoT-based patient monitoring system** can be considered a project with:

- Objective – Develop a real-time patient monitoring system
- Duration – 12 months
- Budget – ₹20 lakh
- Team – Biomedical engineers, software engineers, clinicians and marketing personnel
- Deliverable – Tested and validated prototype/product

#### Major elements of project definition

1. **Project title**
2. **Problem statement**
3. **Objectives**
4. **Scope**
5. **Deliverables**
6. **Target customers**
7. **Timeline**
8. **Budget**
9. **Resources**
10. **Quality requirements**
11. **Regulatory requirements**
12. **Risk management**

#### Importance

A clear project definition helps the team understand **what has to be developed, why it is required, who will use it, how much it will cost, and when it should be completed**.

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**2. Financial Targets**

Financial targets are the **specific financial goals established for a product or project**.

They help determine whether the product is commercially viable.

**Important financial targets**

**1. Development budget**

Maximum amount allocated for product development.

**2. Manufacturing cost**

Cost required to manufacture one unit.

**3. Selling price**

Price at which the product is sold.

**4. Revenue target**

Expected income from product sales.

**5. Profit target**

Expected profit from the product.

**6. Break-even point**

The sales level at which total revenue equals total cost.

**7. Return on Investment (ROI)**

Measures the return generated from an investment.

$$\text{ROI} = (\text{Net Return} / \text{Investment}) \times 100$$

**Example**

Suppose:

- Manufacturing cost = ₹8,000/device
- Selling price = ₹15,000/device
- Expected sales = 1,000 devices

The company can estimate its revenue, contribution margin, operating costs, and expected profit before deciding whether to proceed with the project.

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### 3. Physician Input

#### Introduction

**Physician input** refers to obtaining information, suggestions, feedback, and clinical requirements from doctors during medical product development.

Physicians are important stakeholders because they understand the **clinical problem, patient requirements, workflow, treatment procedures, and limitations of existing technologies**.

#### Importance of Physician Input

##### 1. Identification of clinical problems

Physicians can identify problems that require technological solutions.

##### Example:

A physician may identify that existing patient monitoring systems provide information too late for early intervention.

##### 2. Defining product requirements

Doctors can specify:

- Required parameters
- Accuracy
- Response time
- Clinical functionality
- Patient population
- Desired alerts

##### 3. Product design

Physician feedback helps engineers develop a device suitable for actual clinical environments.

##### 4. Usability

Doctors can evaluate:

- User interface
- Display
- Controls
- Workflow
- Ease of operation

##### 5. Clinical validation

Physicians may participate in clinical evaluation and provide feedback regarding clinical performance.

##### 6. Risk identification

Doctors can identify possible clinical hazards and inappropriate uses.

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#### **7. Product acceptance**

A product developed with appropriate clinical input is more likely to be accepted by healthcare professionals.

##### **Methods of Obtaining Physician Input**

- Interviews
- Questionnaires
- Focus groups
- Clinical observations
- Product demonstrations
- Advisory panels
- Prototype evaluation
- Clinical studies

##### **Example**

For a **smart prosthetic limb**, engineers may develop the sensors and control system, while physicians and rehabilitation professionals can provide input regarding:

- Patient comfort
- Gait requirements
- Safety
- Socket fit
- Rehabilitation needs
- Clinical usability

#### **4. FDA Approval**

##### **Introduction**

The **U.S. Food and Drug Administration (FDA)** regulates medical devices marketed in the United States. FDA's Center for Devices and Radiological Health (CDRH) regulates firms that manufacture, repack, relabel, or import medical devices for the U.S. market.

An important point for students is that **not every medical device receives an "FDA approval."** Depending on the device and regulatory pathway, the product may be **510(k) cleared, De Novo classified, PMA approved, or exempt from a premarket submission.**

##### **Objectives of FDA Regulation**

FDA regulation aims to ensure that medical devices:

- Are reasonably safe
- Are effective for their intended use

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- Meet applicable regulatory requirements
- Are appropriately manufactured
- Have appropriate labeling
- Continue to be monitored after marketing

#### **Step 1 – Identify the Intended Use**

The first step is to clearly define:

- What the device does
- Who will use it
- Which patients will use it
- What disease/condition it addresses
- Whether it diagnoses, treats, prevents, monitors, or supports a condition

The intended use and indications for use are important because they influence the device's classification and regulatory pathway.

#### **Step 2 – Determine Device Classification**

FDA generally classifies medical devices into:

##### **Class I – Low risk**

Generally subject to **General Controls**.

Examples may include certain low-risk devices.

##### **Class II – Moderate risk**

Subject to **General Controls + Special Controls**.

##### **Class III – High risk**

Subject to **General Controls + Premarket Approval (PMA)** in applicable cases.

FDA states that classification is risk-based, with Class I generally representing the lowest risk and Class III the highest.

| <b>Class</b> | <b>Risk</b> | <b>Main Controls</b>                       |
|--------------|-------------|--------------------------------------------|
| Class I      | Low         | General Controls                           |
| Class II     | Moderate    | General + Special Controls                 |
| Class III    | High        | General Controls + PMA in applicable cases |